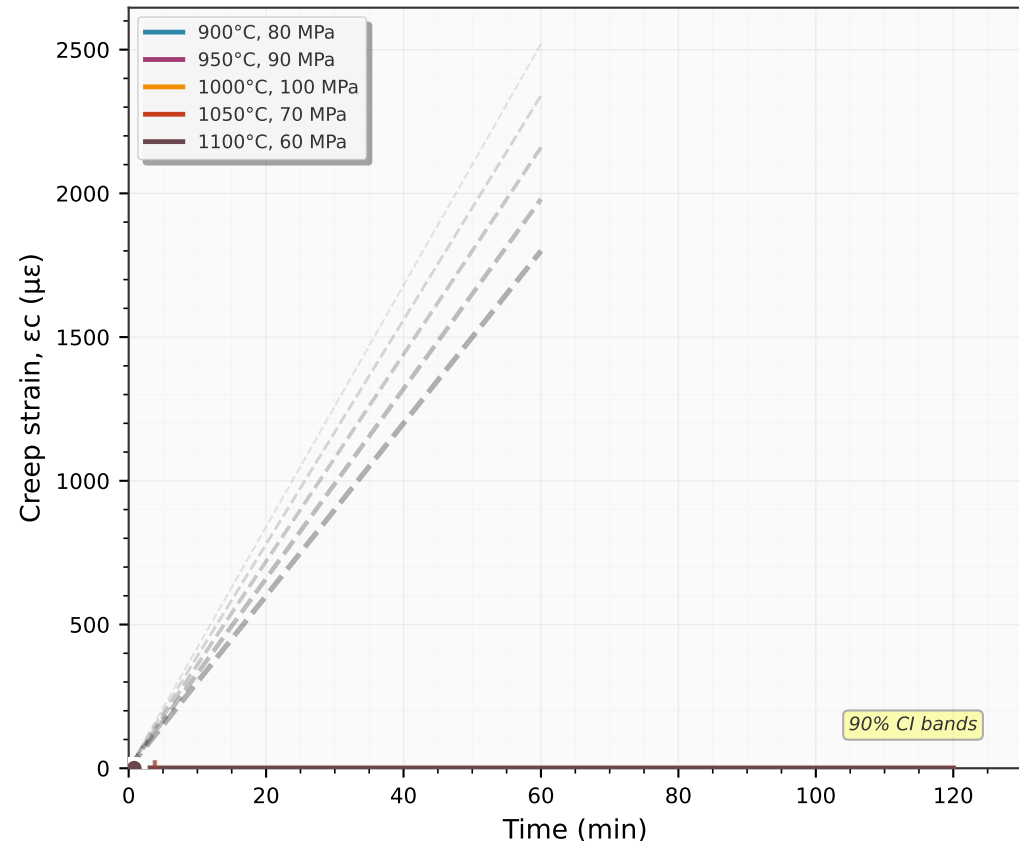


# Advanced Creep-Damage Analysis: Multi-Scale Validation

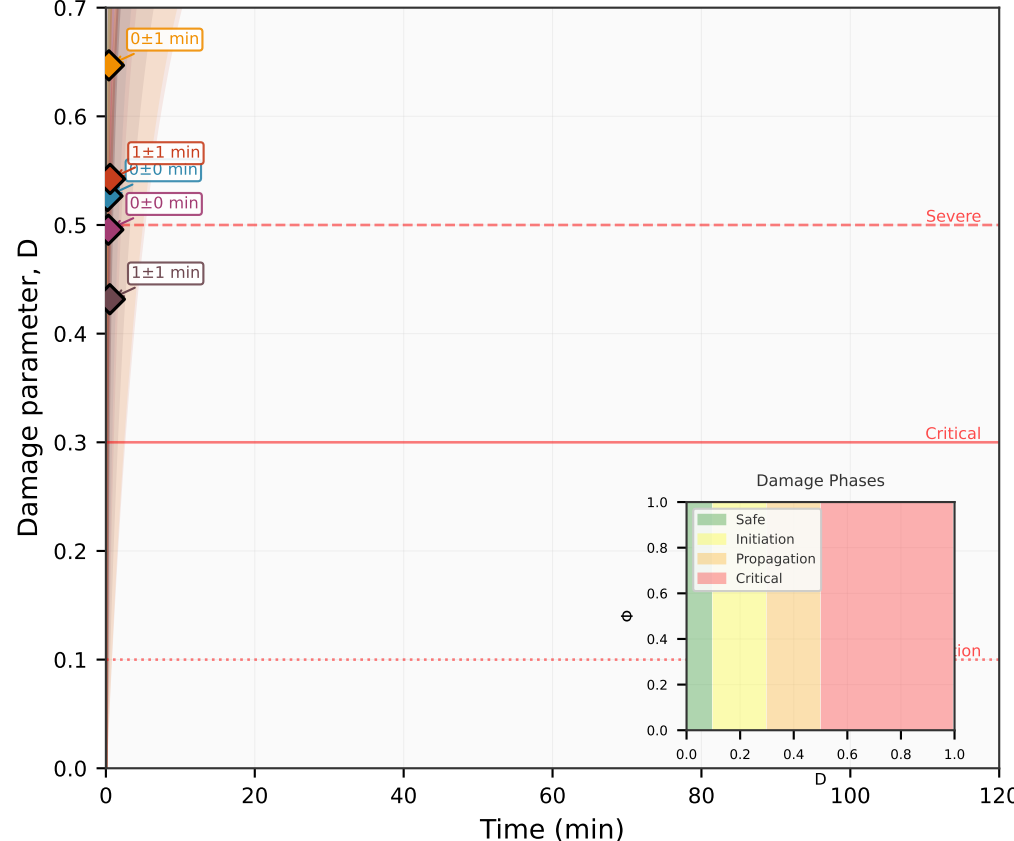
A

Creep Evolution with Confidence Bands



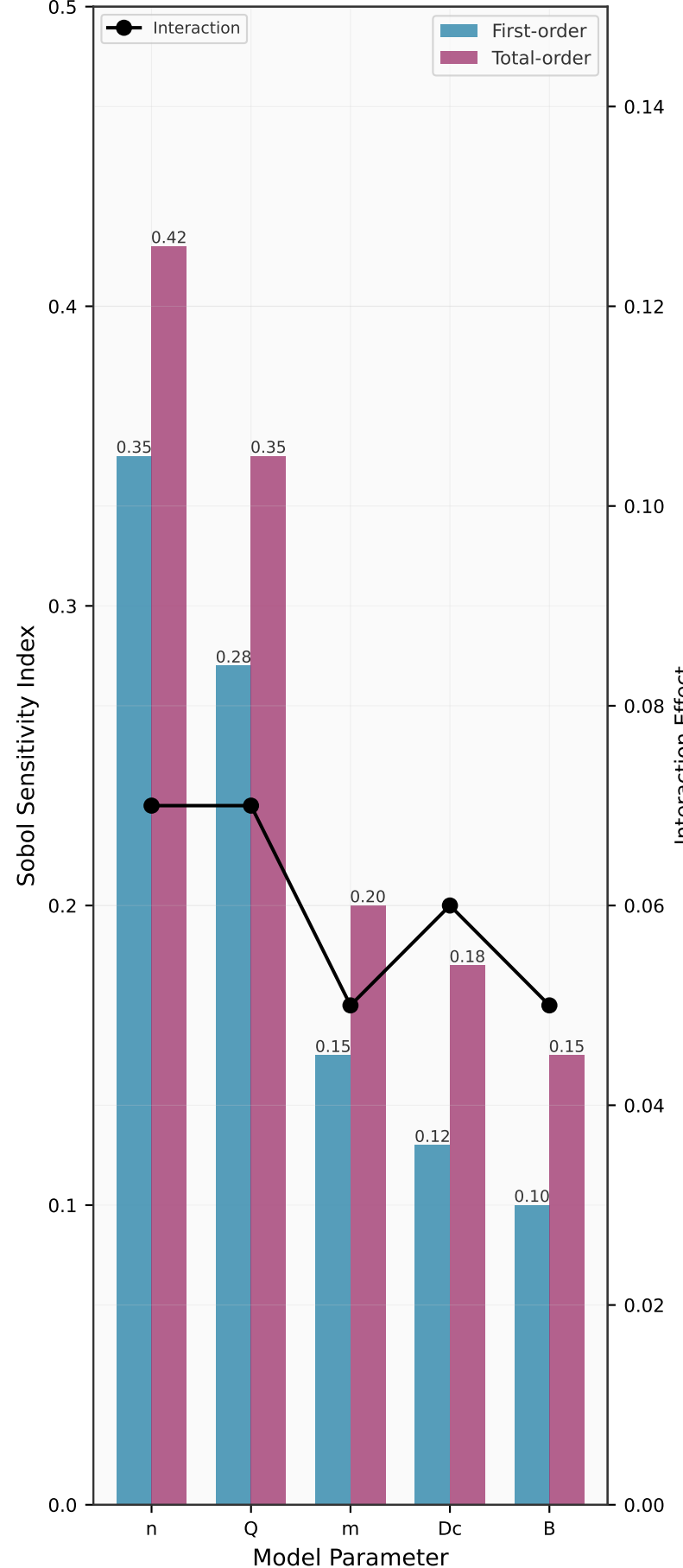
B

Damage Evolution with Phase Transitions



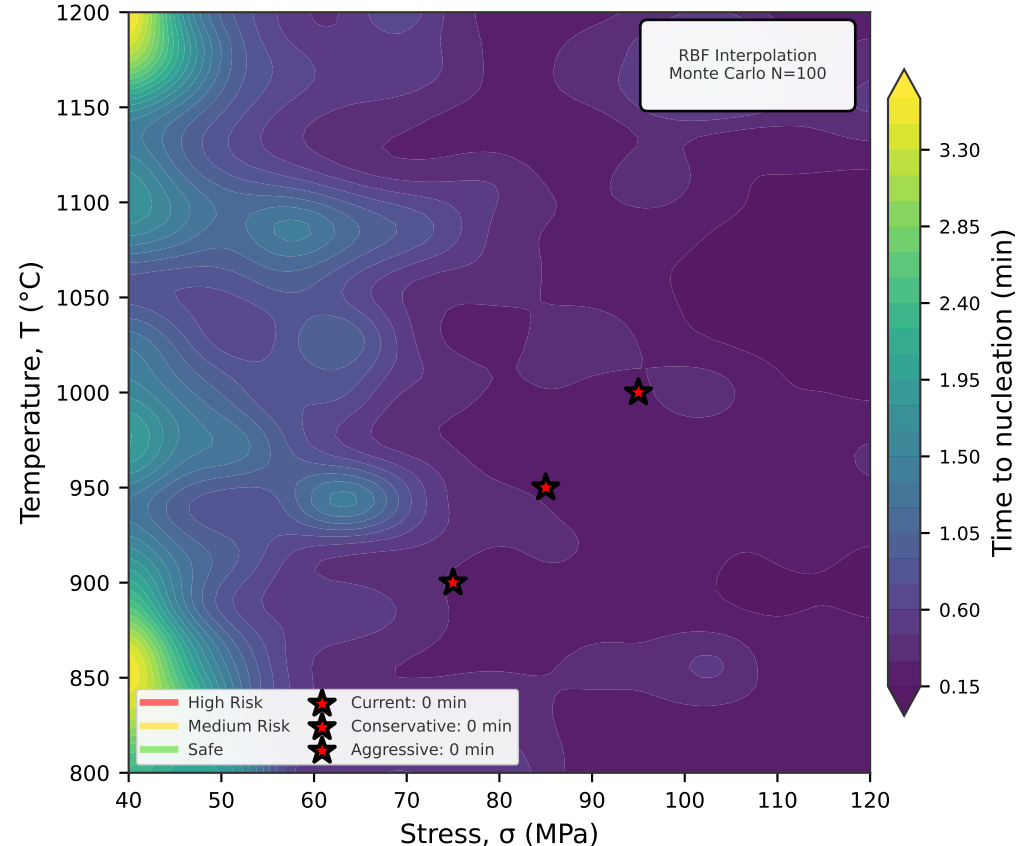
E

Global Sensitivity Analysis



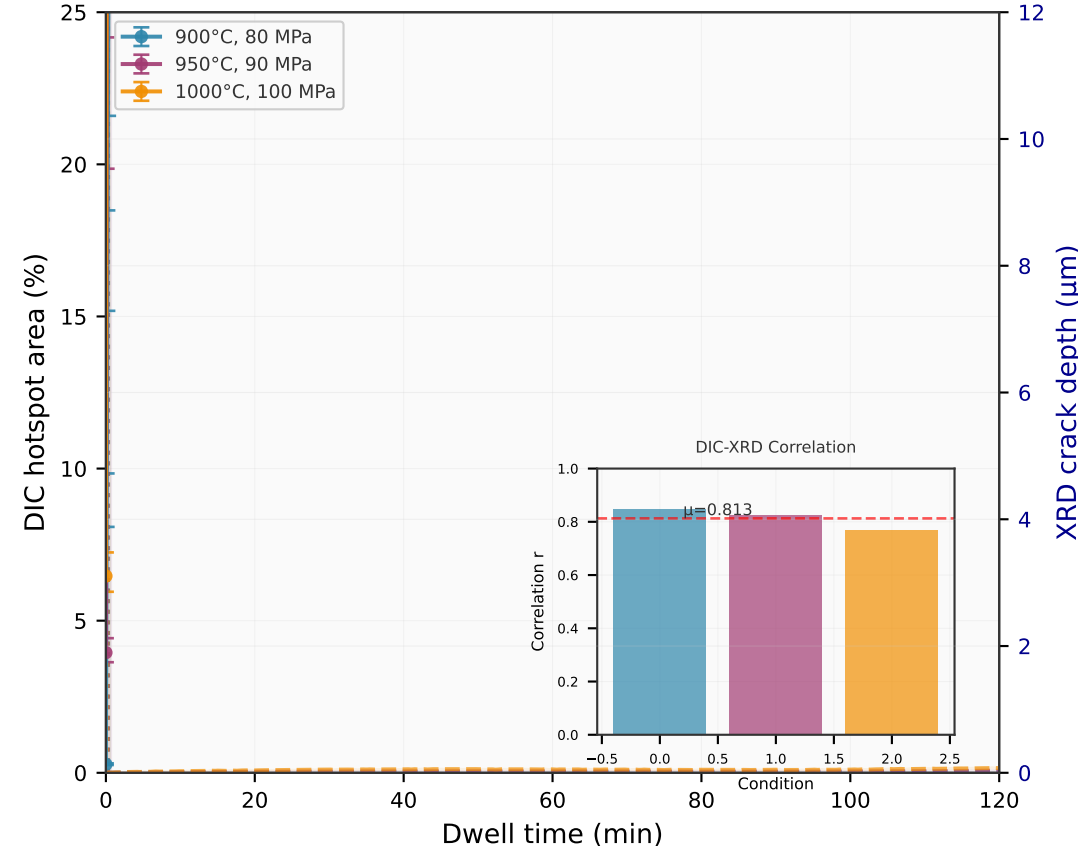
C

3D Hazard Map with Operating Envelope



D

Multi-Modal Experimental Validation



Model: Norton-Bailey/Kachanov-Rabotnov | Confidence: 90% CI | Monte Carlo: N=100

Sobol indices quantify parameter influence on model variance